Faculty of engineering

Department of ECE

Analog Electronic Lab

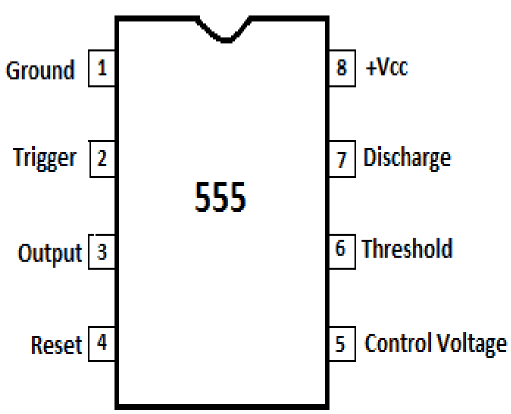
Experiment Name	Timer 555
Student Name	
Grade (10%)	

Objectives

In this Lecture you will design multivibrator circuits that provide signals with particular waveforms.

➞ Astable multivibrator

1. **Objective:** • Design an astable multivibrator to meet a set of specifications.
2. **Specifications:** : The circuit with the configuration shown in Figure below is to be designed to produce frequency of oscillation is ***fo* = 600 Hz and duty cycle is 66 percent ,Then connected your design circuit to load (LED connected in a series with a resistance of 470 Ω).**
3. **Design Approach**: The astable multivibrator configuration to be designed is shown in Figure 1.
4. **Choices:** Timer 555 are assumed to be available, choose +vcc = 9v.
5. **Solution (Analysis):** Considering the circuit in Figure 1, we have let c = 0.1 uf
   - frequency of oscillation is $f = \frac{1}{T} = \frac{1}{T_C + T_D} = \frac{1}{0.693(R_A + 2R_B)C}$
   - duty cycle is $\frac{R_A + R_B}{R_A + 2R_B} \times 100\%$



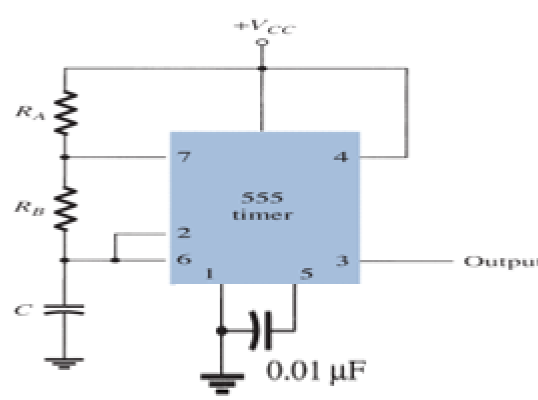


Figure 1: astable multivibrator for the design application

Write your comments.

What happens to the frequency of the waveform if Vcc is increased?

➞ <u>**monastabe multivibrator**</u>

1. **Objective:** • Design an astable multivibrator to meet a set of specifications.
2. **Specifications:** The circuit with the configuration shown in Figure below is to be designed to produce an output pulse width approximately 1m., **Then connected your design circuit to load (LED connected in a series with a resistance of 470 Ω)**
3. **Design Approach**: The monastabe multivibrator configuration to be designed is shown in Figure 1.
4. **Choices:** Timer 555 are assumed to be available, <u>choose +vcc = 9v</u>.

5. **Solution (Analysis):** Considering the circuit in Figure 1, we have

● The width of the output pulse $T = RC\ \ln(3) = 1.1\ RC$

If we set threshold voltage $v(t) = (\frac{2}{3})V^{+}$

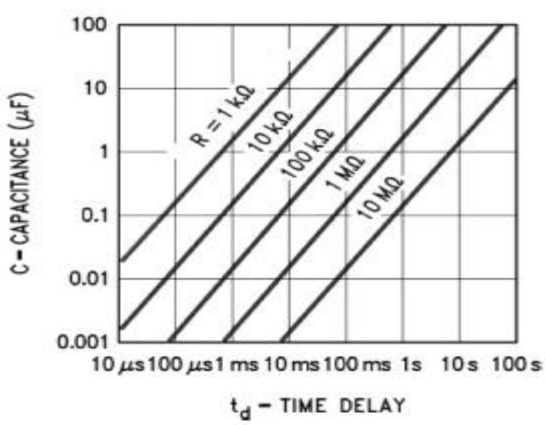


Chart 1

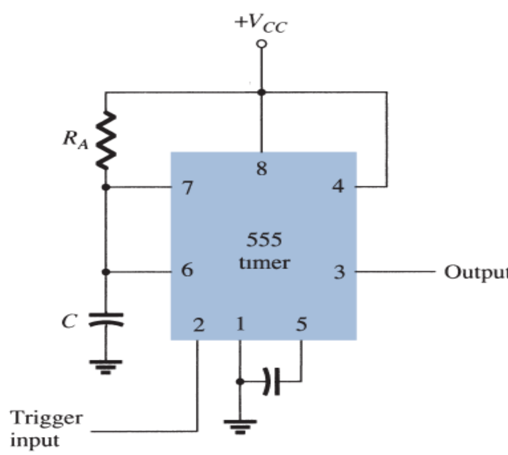
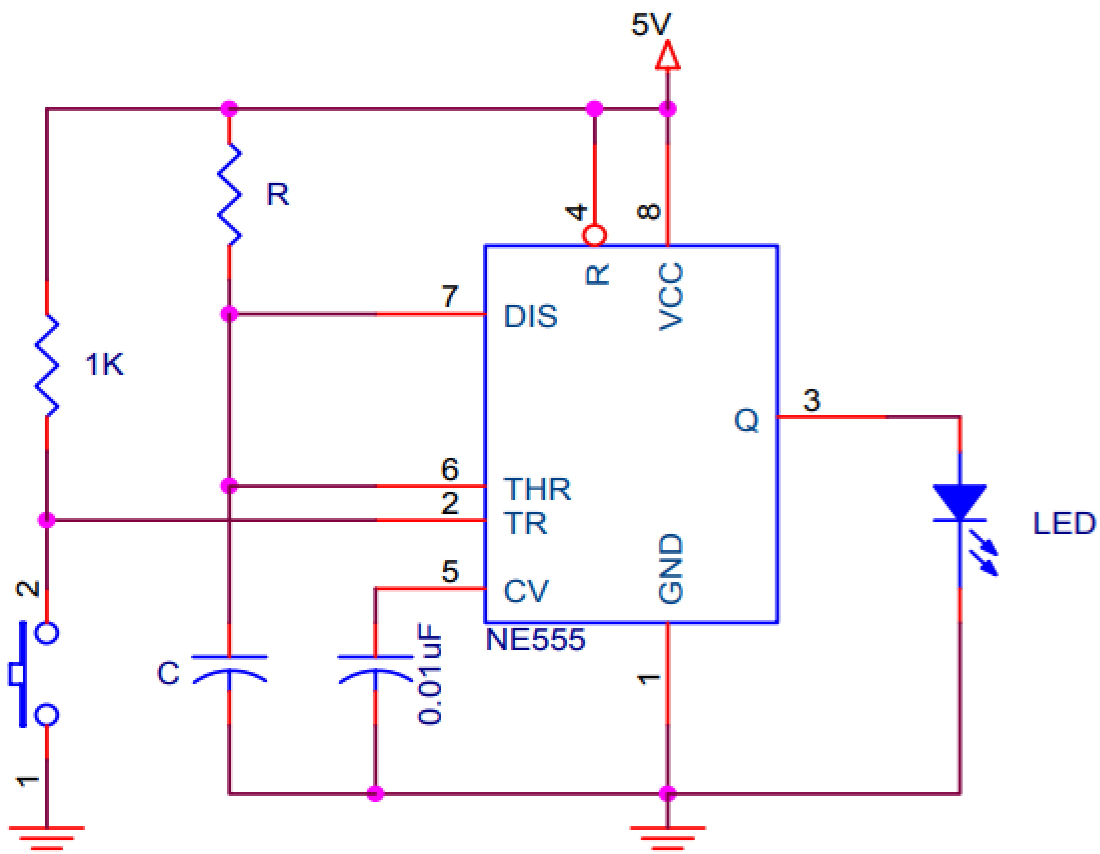


Figure 1: astable multivibrator for the design application

● Use the chart 1 to determine values for R and C to get a pulse that is 1m seconds.

● Use button create a negative pulse to trigger the circuit. When the button is on, the timer will start to charge the capacitor and output goes to high. Once the capacitor reaches a threshold level (2/3 of Vcc), the timer discharges the capacitor and output goes to low


5V
R
4
8
7
DIS
R
VCC
1K
Q
3
6
THR
2
TR
LED
5
CV
GND
2
NE555
C
0.01uF
1
1